WASP-12b

R-1705

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_191013` · created 2026-08-02T14:47:03+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458769.87994 +/- 0.00064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.011
Transit depth (Rp/Rs)^2	1.38 +/- 0.26 [%]
Orbital Inclination (inc)	78.1 +/- 2.51
Airmass coefficient 1 (a1)	1.0373 +/- 0.0096
Airmass coefficient 2 (a2)	-0.0315 +/- 0.0021
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	#1 - [112, 81]
Optimal Method	PSF photometry
Transit Duration (day)	0.108 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.117 ± 0.011 vs 0.1178 (deviation 0.07σ)
Duration fit vs published	0.108 d vs 0.125 d (deviation 1.55σ)
Mid-time O–C	1.2 min
Depth SNR	10.6
Residual scatter	0.92 %

Light curve

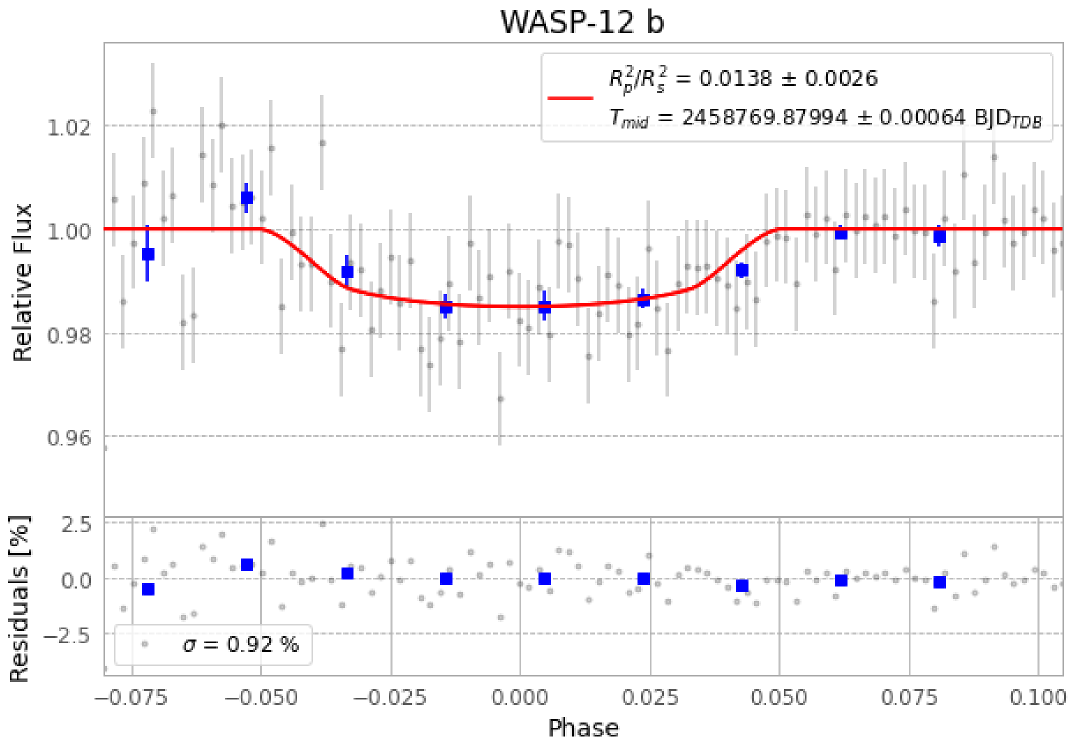

Figure 1 — FinalLightCurve_WASP-12 b_2019-10-12.png (SHA-256 e05135b0abf4117e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [224, 99], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5b74823ad61c6142…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

98 FITS frames (64 MB), `WASP-121910013065712.FITS` → `WASP-121910013114812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-191013.log (b5135c5ac3f9…), FinalLightCurve_WASP-12 b_2019-10-12.png (e05135b0abf4…), FinalLightCurve_WASP-12 b_2019-10-12.csv (03614da7c2df…)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-02 14:47Z.